

Durability of FRP rods for concrete structures

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Abstract

Over the last two decades fiber-reinforced polymer (FRP) rods have emerged as one of the most promising and affordable solutions to the corrosion problems of steel reinforcement in structural concrete. The application of FRP rods in new or damaged structures requires the development of design equations that must take into account the mechanical properties and the durability properties of FRP products.

Concerns still remain about the structural behavior of FRP materials under severe environmental and load conditions for long-time exposures. In the case of glass FRP rods, it is the high pH of the pore water solution created during the hydration of the concrete that may cause the chemical attack of the fibers.

In this study, an effort was made to develop an experimental protocol to study the effects of accelerated aging on FRP rods. The physico-mechanical properties of five types of carbon and glass FRP rods were investigated. FRP specimens were subjected to alkaline simulated concrete pore solution and environmental agents including freeze-thaw, high relative humidity, high temperature and ultraviolet (UV) radiations. Mechanical and physical tests were used to measure the retained properties and to observe the causes of damage and strength reduction. The experimental data showed that resin properties may strongly influence the durability of FRP reinforcement, environmental combined cycles did not take to significant damage of conditioned rod-specimens, GFRP rods are sensitive to alkaline attack when resin does not provide adequate protection to fibers.

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1. Introduction

Beyond the cost issues, the most significant technical obstacle preventing the extended use of FRP materials in civil engineering is a lack of long-term and durability performance data comparable to that available for traditional construction materials. Although there have been numerous studies on creep, stress corrosion, fatigue, environmental fatigue, chemical and physical aging, and natural weathering of composites, most of these studies are not related to civil engineering applications. The importance of these results is relatively low if one considers that composites used for aerospace or naval application are fabricated and assembled under tem-

perature, pressure, and void-content control, while the installation of FRP in construction takes place in-situ. In order to expand the use of FRP in civil structures, relevant durability data must be available for the building codes and standards writing authorities.

The following different damage mechanisms are distinguished in order to classify performance related to the long-term behavior of FRP rods used in civil engineering:

- Fluid absorption and influence on physical and mechanical properties.
- Creep and stress relaxation.
- Fatigue and environmental fatigue damage.
- Weathering.

All these mechanisms can be considered as a consequence of the attack by external agents including:

- Moisture and aqueous solutions.
- Alkaline environment.

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Nomenclature

$M_t(\%)$ percentage of fluid content at time t (g/g%)
 W_d weight of the dry specimen at time $t = 0$ (g)

W weight of the moist specimen at time t (g)

- Thermal effects (freeze-thaw cycling, high temperatures).
- Fatigue loads.
- Ultra-violet (UV) radiation.
- Fire.

There is strong evidence that the rate of degradation of polymer composites exposed to fluid environment is related to the rate of sorption of the fluid [1]. Thus, an understanding of the diffusion process, as well as factors that influence it, is crucial in assessing the state of the material. Theoretical treatment of the diffusion problem can be traced back to the work of previous researchers [2,3]. It can be briefly summarized that the sorption behavior of a polymer or polymer composite depends on: type of fluid, fluid concentration, temperature, applied stress, damage status, chemical structure of the matrix and fiber/matrix interface [4,5].

The effects of moisture or alkaline solution sorption on Glass FRP (GFRP) rods and laminates depend on the mentioned variables and may produce a loss in strength and stiffness [6,7]. The study of the alkaline attack has particular importance in construction applications. FRP rods are immersed in a cementitious environment; this condition was found to be aggressive for GFRP, due to the high pH level (pH 13.5) of the pore water solutions and presence of alkaline ions. The alkaline solution produces, when it can penetrate the composite, an embrittlement of the glass fibers and a damage at the fiber resin interface level by chemical attack and growth of hydration products. These effects lead to a loss in tensile strength and interlaminar transverse properties [8–10]. Different tests were conducted using solutions with high pH, but it has been clearly shown that degradation of fibers is not merely due to the high pH levels, but rather a combination of hydroxylation products (due to the presence of Na^+ and K^+ ions), and presence of moisture and high pH that act as catalysts [8]. Although the use of a polymer matrix as a binder around the glass filaments provides protection to the fiber, there is still concern related to the migration of pH solutions and alkali salts through the resin to the fiber surface. This migration is also enhanced by the presence of tensile stresses, since the fracture toughness of the resin is low. Hence, in the stressed FRP reinforcement, micro cracks in thin outer matrix skin may arise, leading to stress corrosion phenomena.

Another concern is related to the effect of the degradation of the external polymer resin that causes a loss of bond in concrete as previously studied by Bank et al. [11].

Other effects that must be investigated regarding long-term behavior are creep and stress relaxation. In fact, since polymers are viscoelastic materials, they exhibit creep and stress relaxation to a great extent [12]. As a result, FRPs are more susceptible to creep than traditional construction materials, especially under the influence of moisture and temperature [13].

Infrastructure systems are exposed to external agents during their life cycle, so FRP mechanical behavior under natural weathering needs to be understood. Different exposure conditions need to be investigated, including UV radiation, freeze-thaw cycles, high relative humidity, aqueous solution exposure, chemical agents, and combinations of previously described conditions, according to the service condition that are expected.

Thermal effects can result in micro cracks at the interface between FRP rods and concrete because of different thermal expansion coefficients. At high temperature, the bond properties decrease strongly [14]. Freeze-thaw cycling can accelerate solution penetration because of crack growth in the matrix that becomes more brittle. Freeze-thaw cycles without the presence of high moisture do not significantly affect the mechanical properties of FRP rods as reported in laboratory studies [15].

UV exposure leads to surface oxidation due to different chemical mechanisms related to the resin type as investigated in previous studies [16,17]. Previous research found that in AFRP composites, both fibers and resin are affected by UV light, so that potentially a dangerous decrease in mechanical properties may result [18].

2. Research significance

FRP bars embedded in concrete for several years are exposed to an alkaline environment with aggressive ions that may cause a loss in mechanical properties. Five types of FRP rods were subjected to alkaline exposure for different times, with an accelerated aging procedure, using high temperature. The mechanical characterization of the rods provided a measure of durability. In addition, this study investigated how the combination of temperature cycles, high moisture and UV radiation can affect the long-term behavior of the FRP rods. Since, the effects of the single environmental agents were investigated in previous studies, this study allows to assess the combined effect which is more representative of structural applications.

A complete protocol to investigate mechanical properties and long-term behavior of FRP rods is presented. Based on experimental results, a contribution in terms of data related to the long-term behavior of FRP in construction is furnished.

3. Experimental program

3.1. Materials

Two types of GFRP rods (one commercial product and an experimental prototype) and three types of commercial carbon FRP (CFRP) rods were investigated using mechanical and physical tests. The properties of the tested rods are illustrated in Table 1. Different fibers and different resins were used in order to see the effects of external agents on the FRP systems. All FRP products were fabricated using pultrusion process. C1 rods were made with high modulus carbon fibers and epoxy vinylester-modified resin. The surface was made rough by sand-coating and carbon fibers strand wound around. C2 and C3 rods were made with high strength carbon fibers. C1 and C2 rods were fabricated by the same manufacturer using epoxy vinylester-modified resin, while epoxy resin was used in C3. The external surface of C2 rods was the same as C1, while C3 rods presented a light-sandblasted surface. G1 rods were an experimental prototype made with E-glass fibers and a

thermoplastic resin, G2 rods were made with E-glass fibers and a polyester resin. The external surface of G1 rods was smooth, while the external surface of G2 rods was made rough by sand-coating and glass fibers strand wound around.

Differential scanning calorimetry (DSC) was used to measure the glass transition temperature (T_g) range of the resins used as matrix. It was found that G1 and C3 presented a T_g range between 85 and 100 °C, while the glass transition range of C1 and C2 was between 75 and 85 °C. The glass transition in polyester resin of G2 was activated at 70 °C as measured by DSC.

3.2. Mechanical and physical tests

Tensile tests, according to provisions of [19], and short beam test (SBT), according to ASTM D4475, were used to study the changes in longitudinal and transverse mechanical properties, respectively.

Tensile test specimens were prepared using grouted steel anchors, in which the pressure developed by an expansive grout was used to grip the rod. PVC caps and steel washers were used to assure the alignment of the rod during anchor installation. The total length of each specimen was 152 cm except for G2 rods, which had a total length of 122 cm.

ASTM D4475 specimens were cut with different lengths resulting in a span/diameter ratio of 3.6 for C2

Table 1
FRP rods properties

Rod	Fibers	Matrix	Diameter ^a (mm)	Surface
C1	Carbon	Epoxy/vinylester	8.26	
C2	Carbon	Epoxy/vinylester	8.00	
C3	Carbon	Epoxy	7.94	
G1 (prototype)	Glass E	Thermoplastic	12.00	
G2	Glass E	Polyester	6.35	

^a Obtained as average values after 12 measurements for each rod.

specimens, 2.2 for G1 specimens and 3.0 for all other rods. The span was chosen in order to avoid flexural effects that could change the desired shear failure mode. The experimental set-up is illustrated in Fig. 1.

The experimental program is summarized in Table 2. The labels G and C refer to GFRP and CFRP, respectively. Totally, 66 tensile tests and 150 ASTM D4475 tests were performed. ASTM D4475 specimens were dried and weighed every 24 h for the first 10 days, then every 96 h until the end of the aging period, to investigate the absorption properties of the rods immersed in simulated concrete pore solution at 60 and at 22 °C.

Scanning electron microscopy (SEM) was also used in order to observe, from a micro structural point of view, the damage in the FRP systems caused by potential aggressive agents.

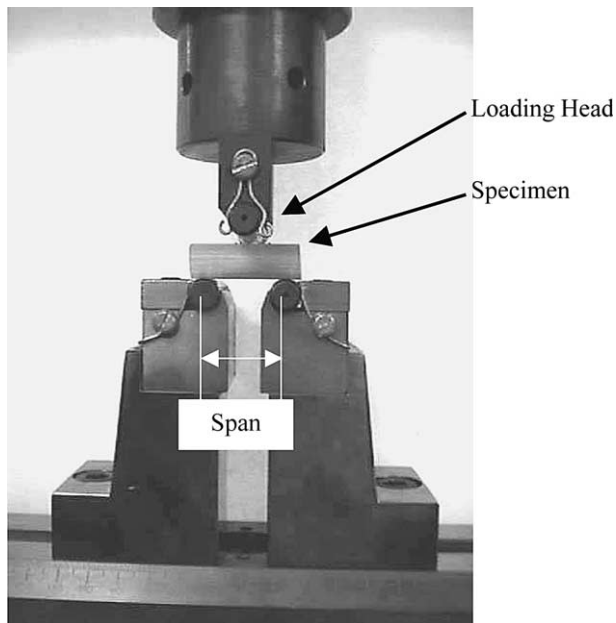


Fig. 1. ASTM D4475 test set-up.

3.3. Accelerated aging

Two aging regimens were used for accelerate conditioning. Twenty tensile specimens and 30 ASTM D4475 specimens were subjected to combined environmental cycles in a controlled chamber: freeze-thaw, high temperature, high relative humidity (HR) cycles and UV radiation. A single complete cycle consisted, first, of 50 freeze-thaw cycles considered as a simulation of winter effects. Here the temperature went from –18 to 4 °C in accordance with the ASTM C666-92 freeze-thaw test standard on concrete. Then high temperature cycles simulated the summer season effects. The temperature cycle was chosen to vary between 16 and 49 °C. In a single environmental cycle, three groups of 50 temperature cycles alternated with three groups of 40 relative humidity (RH) cycles. The first RH cycles started after 50 high temperature cycles to simulate rain during summer days. In the first cycle, RH varied between 60% and 100% at a constant temperature of 16 °C to simulate the humidity and rain during nighttime. In the second group of cycles, RH varied between 60% and 100% at a constant temperature of 27 °C to simulate humidity and rain during daytime. In the third group of RH cycles, it was assumed that as soon as it starts raining, the heat from the concrete may escape causing the nearby air temperature to actually rise to a higher level, so RH varied from 60% to 100% at a constant temperature of 38 °C (100 °F). During high temperature and high RH cycles, the rods were exposed to UV radiation. Totally, each specimen was exposed to 200 freeze-thaw cycles, 480 humidity cycles and 600 high temperature cycles as summarized in Table 3. The environmental cycles are plotted in Fig. 2.

Twenty-four tensile specimens and 90 SBT specimens were subjected to alkali accelerated-exposure. The specimens were immersed in a solution of distilled water containing 0.16% Ca(OH)_2 + 1% Na(OH) + 1.4% K(OH) by weight. The pH factor measured 13.0 before and 12.6 after the period of exposure. It is well known that pH levels may be higher or lower than 12.6 in real

Table 2
No. of specimens tested in the experimental program

Specimens	Control	Alkaline exposure			Environmental cycles
		21 days @ 60 °C	42 days @ 60 °C	42 days @ 22 °C	
G1 tensile	5	3	3	–	4
G2 tensile	5	3	3	–	4
C1 tensile	4	3	3	–	4
C2 tensile	4	–	–	–	4
C3 tensile	4	3	3	–	4
G1 SBT	6	6	6	6	6
G2 SBT	6	6	6	6	6
C1 SBT	6	6	6	6	6
C2 SBT	6	6	6	6	6
C3 SBT	6	6	6	6	6

Table 3
Total environmental exposure

Cycles	Temperature range (°C)	Total number	UV exposure
Freeze-thaw	−18;4	200	No
High temperature	16;49	600	Yes
High RH (60–100%)	16	160	Yes
High RH (60–100%)	27	160	Yes
High RH (60–100%)	38	160	Yes

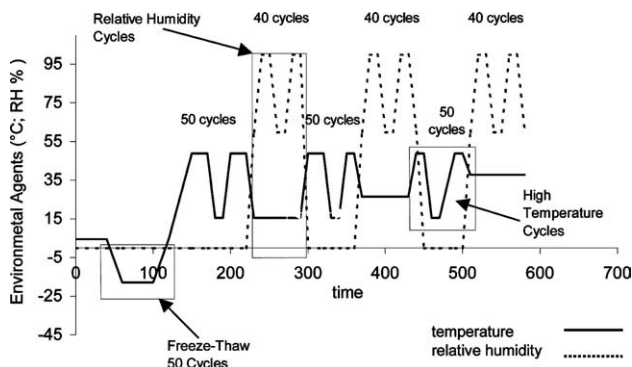


Fig. 2. Environmental cycles.

concrete pore water. This value was taken as an average condition even if this is not the most conservative choice. Previous researches showed how temperature influences the sorption and diffusive properties of alkaline solutions in glass fiber reinforced concrete, comparing natural aging and accelerated test results [20]. Even if these studies were not related to FRP rods for concrete applications, they were considered as a useful reference in order to quantify the acceleration effects due to temperature. A temperature of 60 °C was chosen to accelerate the exposure effects. The potential relationship between the duration of alkali immersion at 60 °C and years of field exposure is considered linear accord-

ing to [20]. Twelve tensile specimens and 30 ASTM D4475 specimens were exposed for 21 days. Twelve tensile specimens and 30 ASTM D4475 specimens were exposed for 42 days. These periods would correspond respectively to 14 and 28 years in a real concrete structure. The correspondence may vary according to the type of concrete. In fact, the pH can change for different cementitious environments as does the presence of K^+ and Na^+ .

As shown in Fig. 3, only a 25 cm section of rod was in contact with the alkaline solution by means of a PVC

Table 4
Tensile test of control specimens

Rod	<i>d</i> (mm)	Ultimate stress (MPa)	Modulus (MPa)	Ultimate strain
G1 test 1	12.00	941	42018	0.0224
G1 test 2		953	42898	0.0222
G1 test 3		951	44547	0.0213
G1 test 4		884	41490	0.0213
G1 test 5		892	41918	0.0213
Mean values		924	42574	0.0217
Standard deviation		34	1216	0.0006
COV		4%	3%	3%
G2 test 1	6.35	365	22750	0.0160
G2 test 2		351	33781	0.0104
G2 test 3		393	34470	0.0114
G2 test 4		407	36745	0.0111
G2 test 5		295	22061	0.0134
Mean values		362	29961	0.0124
Standard deviation		44	6988	0.0023
COV		12%	23%	18%
C1 test 1	8.26	2566	125165	0.0205
C1 test 2		2410	127300	0.0189
C1 test 3		2212	129003	0.0171
C1 test 4		2415	133729	0.0181
Mean values		2401	128799	0.0187
Standard deviation		145	3642	0.0014
COV		6%	3%	8%
C2 test 1	8.00	1900	106780	0.0178
C2 test 2		1878	110422	0.0170
C2 test 3		2251	107628	0.0209
C2 test 4		1946	100200	0.0194
Mean values		1994	106258	0.0188
Standard deviation		174	4328	0.0017
COV		9%	4%	9%
C3 test 1	7.94	1010	110856	0.0091
C3 test 2		1015	104113	0.0097
C3 test 3		1012	106870	0.0097
C3 test 4		1015	110993	0.0091
Mean values		1013	108208	0.0094
Standard deviation		2	3390	0.0004
COV		0%	3%	4%

1 ksi = 6.894 MPa; 1 in. = 25.4 mm.

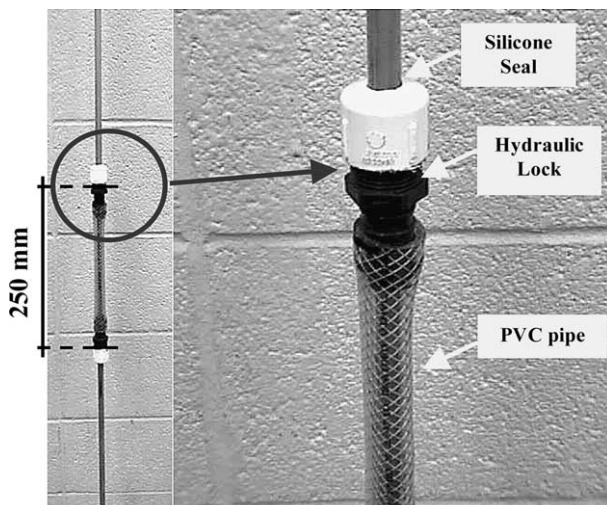


Fig. 3. Alkali exposure set-up.

Table 5
Tensile test after alkali exposure (21 days @ $T = 60^\circ\text{C}$)

Rod	d (mm)	Ultimate stress (MPa)	Modulus (MPa)	Ultimate strain
G1 test 1	12.00	927	41571	0.0223
G1 test 2		939	38193	0.0246
G1 test 3		907	41709	0.0218
Mean values		924	40491	0.0229
Standard deviation		16	1991	0.0015
COV		2%	5%	7%
G2 test 1	6.35	220	26197	0.0084
G2 test 2		276	33712	0.0082
G2 test 3		260	33608	0.0077
Mean values		252	31172	0.0081
Standard deviation		29	4309	0.0004
COV		11%	14%	4%
C1 test 1	8.26	2490	121865	0.0204
C1 test 2		2204	127760	0.0172
C1 test 3		2455	128704	0.0191
Mean values		2383	126110	0.0189
Standard deviation		156	3706	0.0016
COV		7%	3%	8%
C3 test 1	7.94	1045	119749	0.0087
C3 test 2		988	108029	0.0091
C3 test 3		1015	108925	0.0093
Mean values		1016	112234	0.0091
Standard deviation		28	6523	0.0003
COV		3%	6%	3%

1 ksi = 6.894 MPa; 1 in. = 25.4 mm.

tube filled with solution. Hydraulic locks and silicone injections were used to ensure a perfect closure during conditioning.

3.4. Diffusivity and gravimetric measurements

Moisture diffuses into polymers to different extents, depending upon a number of molecular and micro-structural aspects: polarity of the molecular structure, degree of crosslinking, degree of crystallinity (in the case of a thermoplastic), presence of residual monomers, and surface properties. For example, glass surfaces usually contain water-attracting species, or catalyzed anhydride cured resins that contains ester linkages are not very resistant to aqueous solutions, especially acidic and alkaline ones that catalyze the hydrolysis process.

In general, the measurements of weight increase at different times is used to obtain quantitative information about diffusion properties of aqueous solutions in resins and FRP systems. In this research, 18 ASTM D4475 specimens for each type of FRP rod were immersed in simulated concrete pore solution for different times and temperatures (see Table 2). They were dried and

Table 6
Tensile test after alkali exposure (42 days @ $T = 60^\circ\text{C}$)

Rod	d (mm)	Ultimate stress (MPa)	Modulus (MPa)	Ultimate strain
G1 test 1	12.00	903	39709	0.0228
G1 test 2		941	43639	0.0216
G1 test 3		941	39503	0.0238
Mean values		928	40950	0.0227
Standard deviation		22	2331	0.0011
COV		2%	6%	5%
G2 test 1	6.35	170	25784	0.0066
G2 test 2		251	33091	0.0076
G2 test 3		224	32402	0.0069
Mean values		215	30426	0.0070
Standard deviation		41	4035	0.0005
COV		19%	13%	7%
C1 test 1	8.26	2188	127677	0.0171
C1 test 2		2054	129194	0.0159
C1 test 3		2386	125030	0.0191
Mean values		2209	127300	0.0174
Standard deviation		167	2107	0.0016
COV		8%	2%	9%
C3 test 1	7.94	1028	123403	0.0083
C3 test 2		1001	111407	0.0090
C3 test 3		1015	114027	0.0089
Mean values		1015	116279	0.0087
Standard deviation		13	6307	0.0004
COV		1%	5%	4%

1 ksi = 6.894 MPa; 1 in. = 25.4 mm.

weighed periodically. The fluid content was measured as follows:

$$M_t(\%) = \frac{W - W_d}{W_d} \cdot 100, \quad (1)$$

where $M_t(\%)$ is the percentage of fluid content at time t , W_d is the weight of the dry specimen at time $t = 0$ and W is the weight of the moist specimen at time t .

If the fluid absorption follows Fick's law, as usually happens in FRP systems, in the first part of the exposure time, the diffusivity appears to be independent to moisture concentration and of its through thickness location.

In some cases, after a certain time the diffusion phenomena do not follow Fickian kinetics, and an equilibrium moisture concentration is not achieved since the resin may be damaged. In these cases, the fluid content is higher and the slope of the weight increase curve versus square root of time increases dramatically. This indicates that the fluid penetrated inside the composite causing potential damage of the fibers when the fluid contains ions that are chemically aggressive for that kind of fiber.

Table 7
Tensile test after environmental exposure

Rod	<i>d</i> (mm)	Ultimate stress (MPa)	Modulus (MPa)	Ultimate strain
G1 test 1	12.00	896	39985	0.0224
G1 test 2		897	41709	0.0215
G1 test 3		982	39916	0.0246
G1 test 4		856	39296	0.0218
Mean values		908	40226	0.0226
Standard deviation		53	1036	0.0014
COV		6%	3%	6%
G2 test 1	6.35	309	24749	0.0125
G2 test 2		421	27576	0.0153
G2 test 3		309	27576	0.0112
G2 test 4		316	29851	0.0106
Mean values		338	27438	0.0124
Standard deviation		55	2089	0.0021
COV		16%	8%	17%
C1 test 1	8.26	2490	125030	0.0199
C1 test 2		2490	126326	0.0197
C1 test 3		2443	123699	0.0197
C1 test 4		2295	128146	0.0179
Mean values		2430	125800	0.0193
Standard deviation		92	1896	0.0009
COV		4%	2%	5%
C2 test 1	8.00	2085	117288	0.0178
C2 test 2		1891	110993	0.0170
C2 test 3		1763	105340	0.0167
C2 test 4		1864	107388	0.0174
Mean values		1901	110252	0.0172
Standard deviation		135	5240	0.0004
COV		7%	5%	3%
C3 test 1	7.94	997	105478	0.0094
C3 test 2		1032	100652	0.0103
C3 test 3		979	100652	0.0097
C3 test 4		1032	107546	0.0096
Mean values		1010	103582	0.0098
Standard deviation		27	2786	0.0004
COV		3%	3%	4%

1 ksi = 6.894 MPa; 1 in. = 25.4 mm.

4. Test results

4.1. Tensile properties

Tensile tests showed an elastic behavior until brittle failure occurred, which is the typical behavior of FRP composites with unidirectional fibers. All specimens failed showing tensile fiber rupture. Tensile properties of control specimens are reported in Table 4. G1 rods showed a tensile strength of 924 MPa with a modulus of 42.6 GPa. G2 rods showed low mechanical properties, due to a low fiber content as seen by SEM investigations. The ultimate tensile strength was 362 MPa with a

modulus of 30 GPa. It was observed that tensile failure of G1 specimens was accompanied by delamination of the fibers, even if the failure was always brittle. CFRP tensile specimens showed catastrophic failure, with a violent release of elastic energy that caused expulsion of the anchorages from the machine frame. C1 rods showed a tensile strength of 2401 MPa with a modulus of 129 GPa. Ultimate strength of C2 was lower than C1 even if the manufacturer indicated high strength fibers in C2, the ultimate strength was 1994 MPa with a modulus of 106 GPa. C3 showed lower properties; the ultimate strength was 1013 MPa with a modulus of 108 GPa.

Different values of residual tensile properties were measured on conditioned rods. Test results related to conditioned specimens are reported and summarized in Tables 5–7.

Alkaline solution exposure dramatically affected G2 rods. In fact, the ultimate strength showed a reduction of 30% after 21 days exposure, and of 40% after 42 days exposure. G1 rods did not show a decrease in mechanical longitudinal properties. Tensile properties measured on conditioned specimens were the same as for control G1 rods. Environmental cycles did not reduce the tensile strength of both GFRP specimens since the same values as control series were measured for G1 and G2 exposed rods.

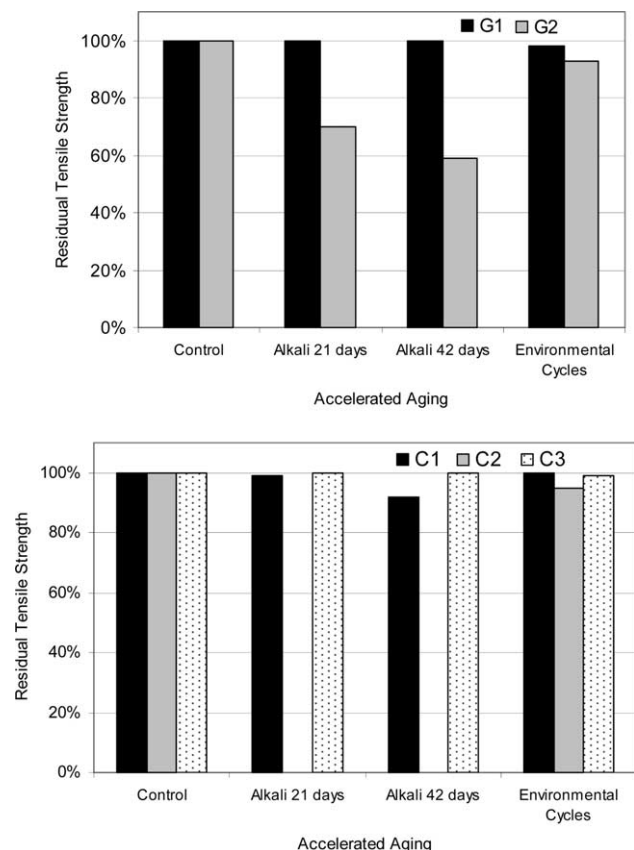


Fig. 4. Residual tensile strength.

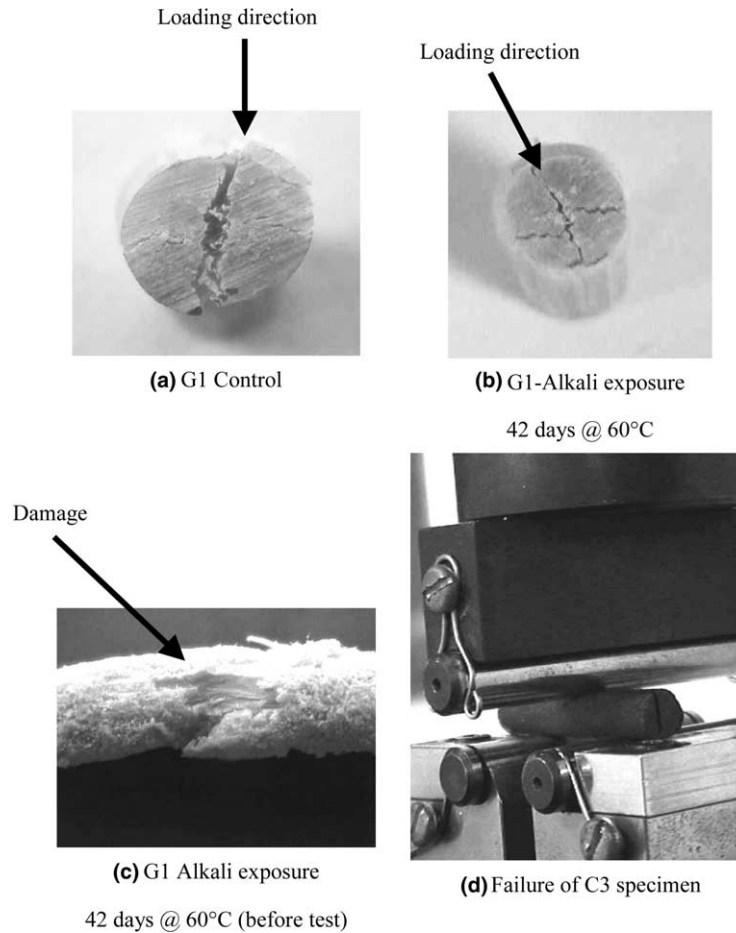


Fig. 5. ASTM D4475 specimens.

CFRP specimens had a good retention of mechanical properties after both conditioning exposures: C1 rods experienced a 8% reduction after 42 days in alkaline solution that could be due to resin degradation more than fiber damages and may be considered within the experimental error. Tensile properties of C3 were not affected by immersion in simulated concrete pore solution.

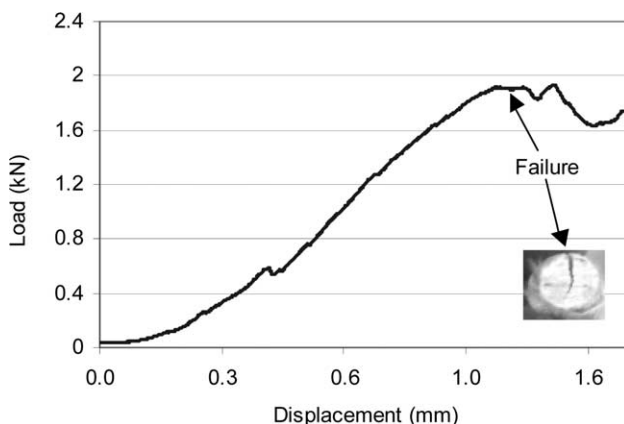


Fig. 6. Load displacement curve for G2 rods in ASTM D4475 test.

In Fig. 4 the residual performance of exposed FRP rods is compared with control properties. It is clear that in all cases, environmental cycles did not cause important reduction of tensile strength.

4.2. Transverse properties (apparent horizontal shear strength)

Transverse properties determined by ASTM D4475 are mostly related to resin properties. Therefore the apparent horizontal shear strength results may furnish a measure of resin damage caused in this case by penetration of fluids. Since aggressive ions present in penetrated fluids may damage the fibers, these results may furnish also indications on possible effects on the longitudinal mechanical properties.

G1 control specimens showed a vertical plane of failure (plane of load) with crack perpendicular to the cross-section of the rod as illustrated in Fig. 5(a). Other cracks were developed in the horizontal direction and were evident after testing of conditioned specimens as illustrated in Fig. 5(b). Therefore conditioned specimens had a combined horizontal and vertical shear failure due to extensive cracking developed under load also along

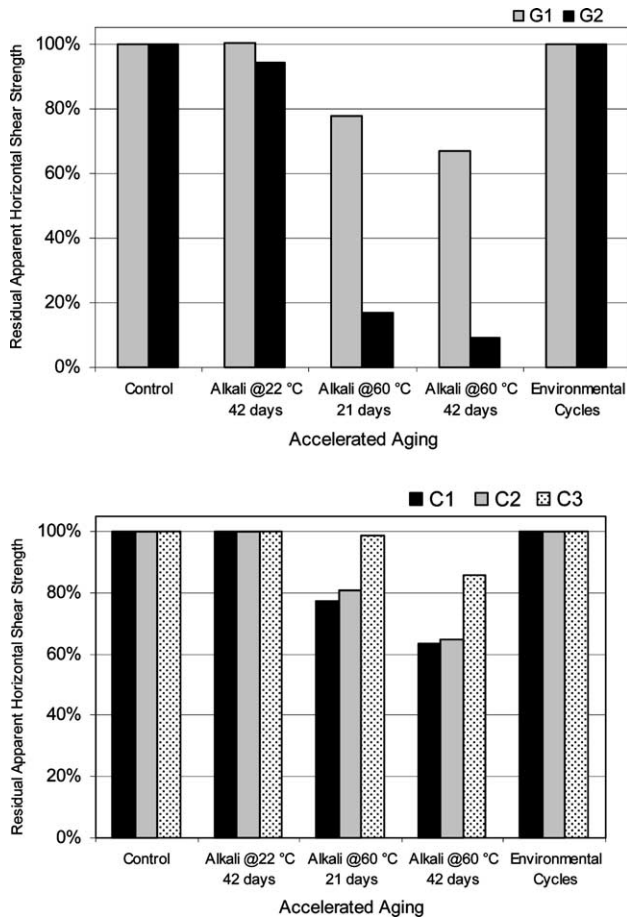


Fig. 7. Residual apparent horizontal shear strength after alkali exposure.

the horizontal plane. The thermoplastic nature of the resin caused a penetration of the loading head into the specimen before the reach of ultimate load.

G2 control specimens showed a load-displacement curve reported in Fig. 6, with a vertical failure shear plane due to the ultimate horizontal shear strength of the polyester resin. Conditioned specimens after 42 days at 60 °C presented extensive damage before testing, due to fluid penetration as illustrated in Fig. 5(c). In these cases, the failure resulted in a complete splitting of the cross-section along different planes.

As expected for epoxy CFRP rods, a neat vertical failure shear plane with crack perpendicular to the transverse cross-section of the rod was clearly evidenced after testing of carbon specimens C1, C2 and C3 under all conditions, as illustrated in Fig. 5(d) for C3.

Epoxy matrix in C3 rods presented unaffected properties, while epoxy vinylester-modified resin in C1 and C2, and thermoplastic resin in G1 prototype were moderately affected by the diffusion of solution. G2 rods were strongly affected; in fact, the residual properties of the polyester matrix decreased almost to zero after conditioning. Cracking patterns in G2 were visible after conditioning, also without microscopy investigations. In Fig. 7 residual apparent horizontal shear strength is illustrated. It can be observed that the effect of temperature was significant. In fact, specimens immersed at room temperature for 42 days did not show any variation from the control ones, while specimens immersed at 60 °C highlighted a reduction in transverse properties.

All specimens subjected to combined environmental cycles were not affected by conditioning with regards to apparent horizontal shear strength. In all cases the residual properties were the same as those measured for control specimens.

4.3. Absorption properties

Gravimetric measurements allowed to investigate the weight increase due to fluid diffusion into FRP rods.

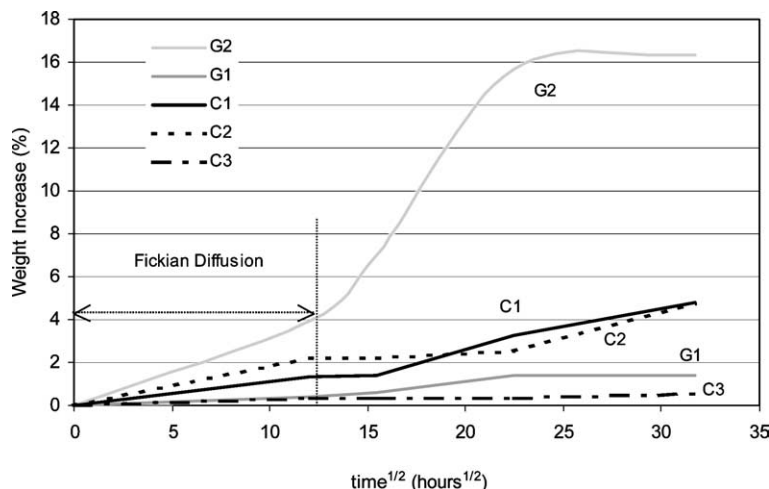


Fig. 8. Absorption behavior of FRP rods.

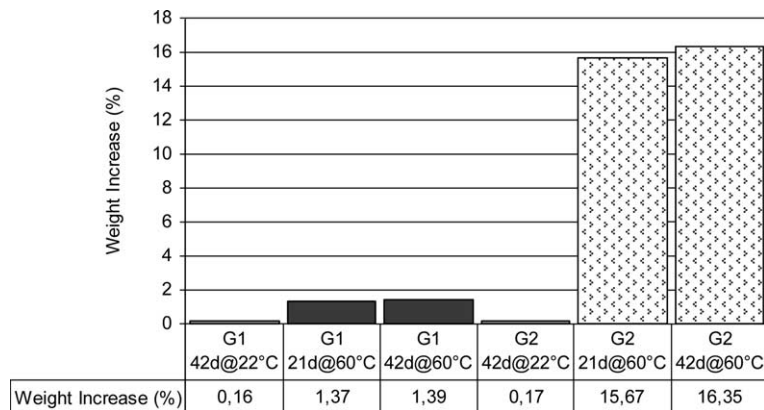


Fig. 9. Weight increase after immersion (GFRP specimens).

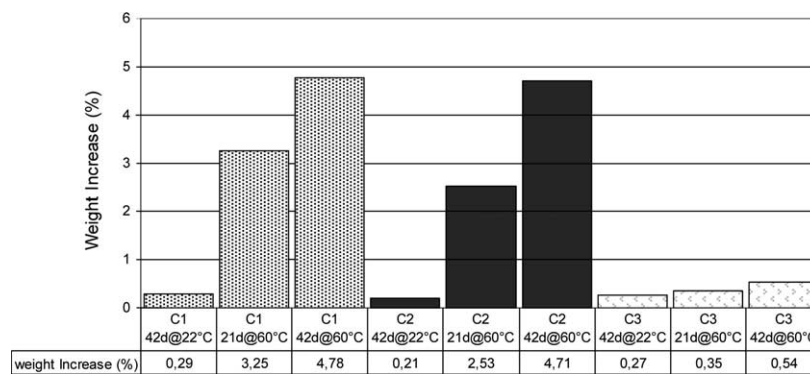


Fig. 10. Weight increase after immersion (CFRP specimens).

This property is related to the permeability of the resin that should furnish a physical coating to the fibers, especially for glass fibers that are chemically sensitive if compared with carbon fibers that are more resistant.

An initial linear diffusion curve was observed for the different resins. In Fig. 8 the diffusion curves are reported for FRP specimens immersed in alkaline solution. A first period of linear diffusion, conforming to Fick's law, can be observed. Then the different amount of damage produced by fluid penetration at 60 °C generated a different rate of absorption that was very high for polyester. Low absorption was measured for epoxy in C3 rods. In this region, the assumption of Fickian diffusion kinetics is not valid. The change of slope in diffusion curves highlighted the changes in absorption mechanisms that are controlled by non linear diffusion kinetics. It can be seen also that in C1 and C2 rods, after 42 days of immersion, the slope of the curves is positive, meaning that fluid penetration did not reach an equilibrium as observed for the other coupons. In Figs. 9 and 10 the weight increase is reported for different immersion conditions. It may be observed that the values of weight increase for specimens immersed at 22 °C was very low if compared with immersion at 60 °C. These

results highlighted the important role of temperature in absorption kinetics of polymeric resins.

4.4. Electronic microscopy (SEM)

Electronic microscopy allowed to understand how the chemical attack occurred after immersion in alkaline solution and which part of the microstructure of the FRP acted as the weak link in the damage development. SEM images, illustrated in Fig. 11, show that the edge of G1 rod is affected by alkali, but the core of the section is not. Fiber damage of the external fibers in G1 is illustrated. The alkaline attack resulted in complete breakage of the external fibers, as reported in Fig. 11(b).

G2 rods were strongly attacked by alkali, and extensive cracking can be observed. Fig. 11(c) highlighted cracks in the matrix and damage in the fiber due to alkaline fluid penetration. Fig. 11(d) shows a SEM observation of the specimen illustrated in Fig. 5(c). The extensive cracking caused by the low properties of the resin in alkaline wet environment is evident. C1 and C3 rods did not present damage at fiber level as illustrated in Fig. 10(f) and (h). C1 rods showed a weak fiber/matrix interface as visible in Fig. 11(e). This defect proba-

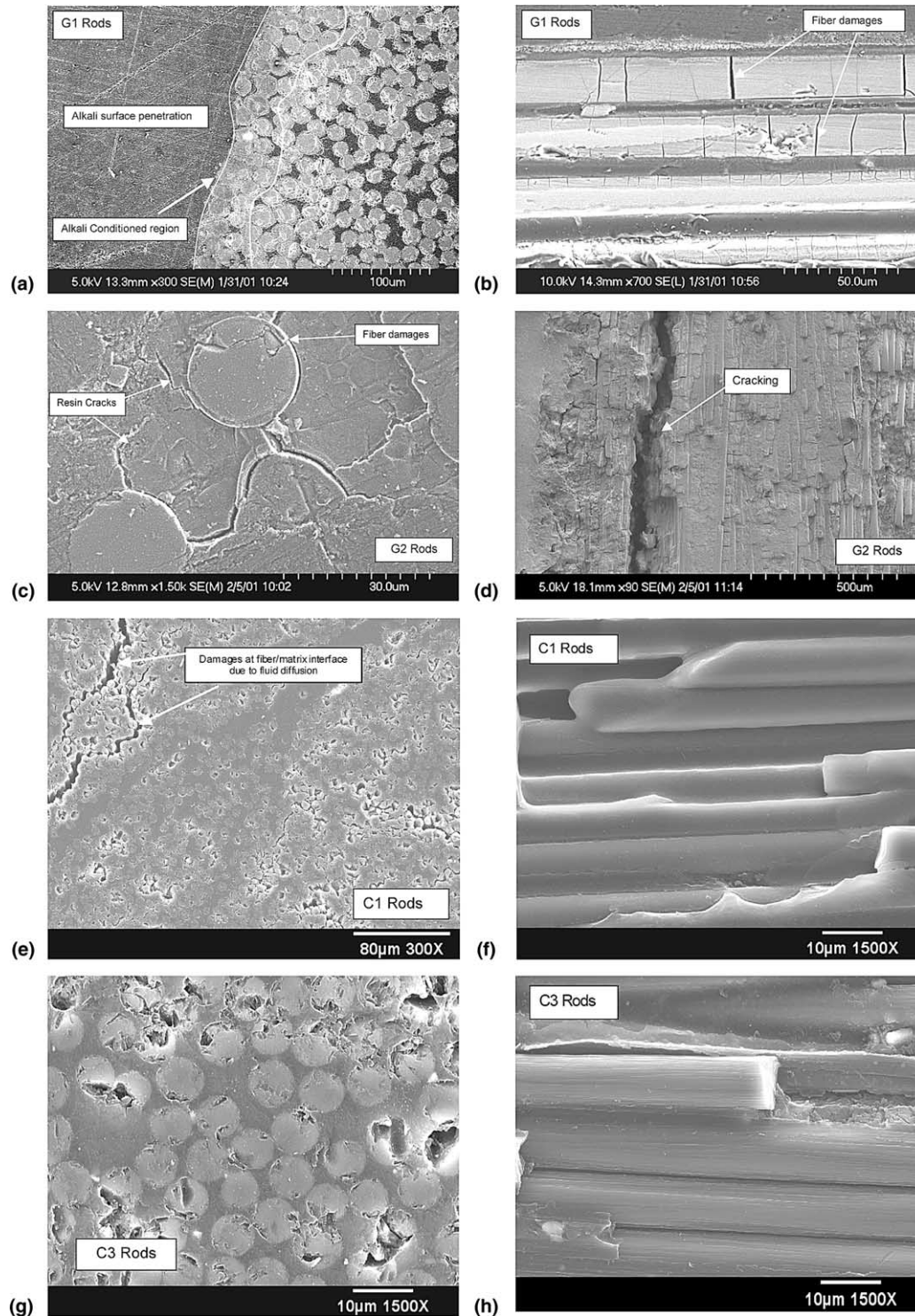


Fig. 11. SEM images of alkali conditioned FRP rods.

bly generated the fluid penetration that resulted in a moderate moisture content after fluid immersion. Damage at fiber level due to cutting are also visible, but they are not related to the chemical or physical attack by external agents.

5. Conclusions

An experimental program was implemented to develop a protocol for mechanical and physical characterization of FRP rods used in construction. Durability

effects due to environmental exposure and immersion in simulated concrete pore solution were studied. A combination of physical and mechanical tests was undertaken. Concluding remarks are summarized as follows:

- Short shear span test according to ASTM D4475 is recommended in order to study resin properties after aging exposure. Results of ASTM D4475 tests showed a reduction in apparent horizontal shear strength of about 30% for C1, C2 and G1 specimens, while the reduction was less than 20% for C3 after 42 days of immersion at 60 °C in simulated concrete pore solution. G1 transverse properties decreased almost to zero after the aging period because of extensive damage in polyester resin.
- Gravimetric measures were in accordance with ASTM D4475 results since higher absorption values corresponded to low transverse mechanical properties after conditioning. Thus gravimetric tests may be recommended after any solution immersion, since the weight increase may furnish, without any other information, a measure of potential degradation of the FRP system.
- Electronic microscopy would be recommended when it is necessary to know the damage mechanisms caused by aggressive agents. Fiber, matrix, and interfacial defects could be well detected after accelerated aging. SEM observations of C3 rods showed the integrity of fiber/matrix interface due to the good quality of the epoxy resin used. In C1 and C2 bars after the conditioning regimen damage at fiber/resin interface was observed in presence of epoxy vinyl-ester-modified resin. The chemical action of alkali ions was evident only in the external layer of G1 specimens, while G2 rods resulted extensively damaged at resin and fiber level as visible also without the aid of microscopy.
- Tensile tests showed that G1 and C3 rods had a retention of tensile properties of 100% after immersion in simulated concrete pore solution and after environmental cycles; C1 and C2 rods showed good durability, the amount of reduction in C1 rods was within the experimental error. Tensile strength of G2 rods decreased to 70% and 59%, if compared to strength of virgin specimens, respectively after 21 and 42 days of immersion in alkaline solution.
- Environmental combined cycles did not produce significant damage in conditioned composites and residual physic-mechanical properties did not show a decrease in all cases.
- Polyester resin did not provide adequate protection for glass fibers that were strongly damaged by alkali ion penetration. According to the results of the present study, the use of the tested polyester resin in GFRPs designed for the construction industry is not recommended.

- Carbon fibers did not show weakness against alkaline solutions, even if fluid penetration that caused crack at fiber/matrix level was observed in C1 and C2 rods.
- The use of thermoplastic resin revealed to be a promising approach in terms of mechanical behavior respect to long-term potential degradation of E-glass fibers.

6. Recommendations for future work

The major limitation of the experimental work presented herein is the absence of stress during the accelerated aging of the rods. Further research should investigate also the mechanical behavior after aging accompanied by stress application.

It was found that the resin properties may strongly influence the durability of FRP reinforcement used in construction. Therefore, future efforts devoted to recommendation of safety design reduction coefficients should be accompanied also by acceptance criteria in terms of constituent materials. It is considered necessary in order to guarantee long-term quality that should help to increase the confidence of designers and contractors for using of FRP reinforcement in civil structures.

Furthermore, the conclusions reached in this study cannot be generalized or taken out of contest because each FRP system has specific constituents and fabrication methods.

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